Type	Prompt	Golden	Generation	
			w/ without addition	w/ Snowballing
HK^+	What color are the stars on an official United States flag?	White	White	Red
factually-correct	A seriema is what type of creature?	Bird	A bird	A bird
HK^-	Which insect does Isle of Wight Disease affect?	BEE	The Whitefly	Mosquito

Table 7: Generated answers using greedy decoding on TriviaQA with Mistral.

Dataset	# Factually correct	# Hallucination (HK^+)	# Do-not-know (HK^-)
TriviaQA-Llama3-WACK	14154	1675	7356
Natural-Questions-Llama3-WACK	5934	1104	14739
TriviaQA-Gemma-WACK	13534	2563	6991
Natural-Questions-Gemma-WACK	6045	1859	13762
TriviaQA-Mistral-WACK	12652	2841	7650
Natural-Questions-Msitral-WACK	5562	1546	14689

Table 8: Dataset labels statistics using Snowballing setting.

Dataset	# Factually correct	# Hallucination (HK^+)	# Do-not-know (HK^-)
TriviaQA-Llama3-WACK	14851	978	7356
Natural-Questions-Llama3-WACK	6059	979	14739
TriviaQA-Gemma-WACK	15418	679	6991
Natural-Questions-Gemma-WACK	7194	710	13762
TriviaQA-Mistral-WACK	14505	988	7650
Natural-Questions-Msitral-WACK	6232	876	14689

Table 9: Dataset label statistics on the Alice-Bob setting.

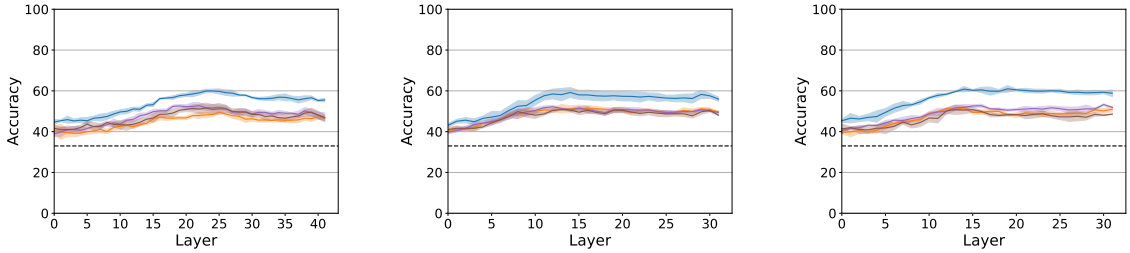
Dataset	# Factually correct	# Hallucination (HK^+)	# Do-not-know (HK^-)
TriviaQA-Llama3-WACK	15062	767	7356
Natural-Questions-Llama3-WACK	6153	885	14739
TriviaQA-Gemma-WACK	15445	652	6991
Natural-Questions-Gemma-WACK	7195	709	13762
TriviaQA-Mistral-WACK	14575	918	7650
Natural-Questions-Msitral-WACK	6218	890	14689

Table 10: Dataset label statistics on the Persona setting.

classification results when distinguishing between the two hallucination types, HK^+ and HK^-. Distinguishing these two types is easier under the snowballing setting (sometimes getting close to 80%), but even under other settings, results are above chance.

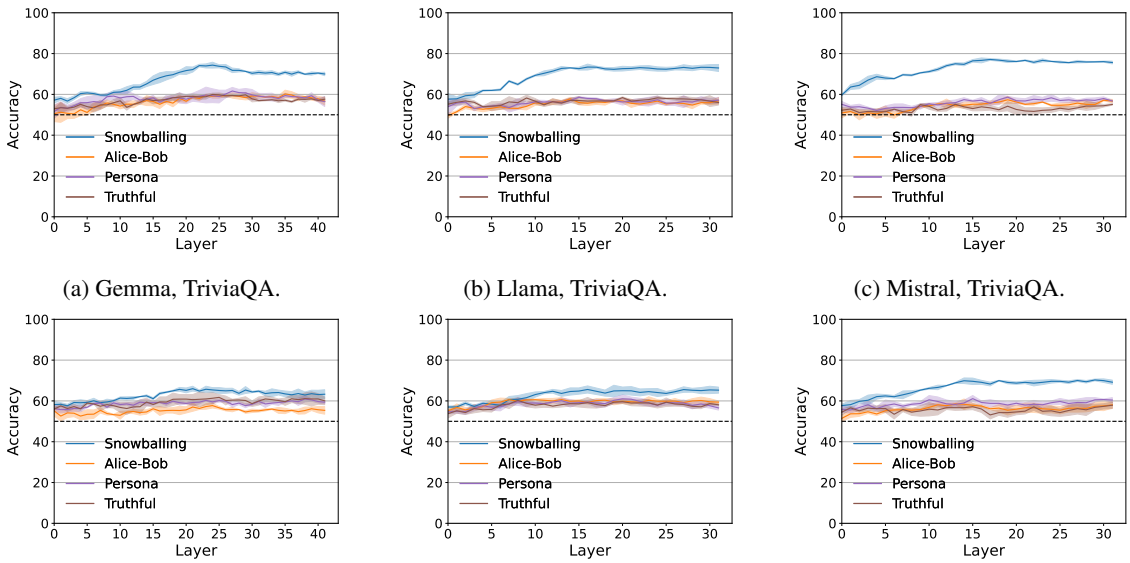
F Additional Results on the Difference between Models Hallucinations

In Section 5.1, we demonstrate that different models possess distinct knowledge and exhibit unique types of hallucinations. Additionally, even within the same model, the hallucinations can vary depending on the



(a) Gemma. (b) Llama. (c) Mistral.

Figure 7: 3-way classification results on Natural Questions into (i) hallucinations caused by lack of knowledge (HK^-), (ii) hallucinations caused despite having knowledge (HK^+), and (iii) factually correct examples. All the detection results are well above the random baseline (dashed line) indicating the possibility of distinguishing HK^+ from HK^-.



(a) Gemma, TriviaQA. (b) Llama, TriviaQA. (c) Mistral, TriviaQA.

(d) Gemma, Natural Questions. (e) Llama, Natural Questions. (f) Mistral, Natural Questions.

Figure 8: Classification results into (i) hallucinations caused by lack of knowledge (HK^-), (ii) hallucinations caused despite having knowledge (HK^+). We can see that all the detection lines are well above the random baseline (dashed line) indicating the possibility of distinguishing HK^+ from HK^-.

Dataset	# Factually correct	# Hallucination (HK^+)	# Do-not-know (HK^-)
TriviaQA-Llama3-WACK	14987	842	7356
Natural-Questions-Llama3-WACK	6203	835	14739
TriviaQA-Gemma-WACK	15458	639	6991
Natural-Questions-Gemma-WACK	7253	651	13762
TriviaQA-Mistral-WACK	14728	765	7650
Natural-Questions-Msitral-WACK	6421	687	14689

Table 11: Dataset label statistics on the Truthful setting.

Model	Data Set	HK^+	HK^-
Gemma	TriviaQA	**15.4 / 20.0 / 19.6 / 7.8**	2.2 / 1.6 / 1.8 / 2.2
	Natural Questions	**17.2 / 23.2 / 19.0 / 11.6**	2.2 / 1.0 / 1.6 / 1.8
Llama	TriviaQA	**18.6 / 26.2 / 16.4 / 11.6**	1.4 / 1.6 / 0.6 / 2.2
	Natural Questions	**16.8 / 25.4 / 20.2 / 18.4**	1.0 / 1.4 / 0.6 / 1.0
Mistral	TriviaQA	**24.6 / 17.4 / 18.6 / 18.2**	1.8 / 3.2 / 1.0 / 3.4
	Natural Questions	**23.6 / 20.0 / 15.4 / 16.4**	1.4 / 2.2 / 0.8 / 1.4

Table 12: Comparison of HK^+ and HK^- mitigation results using alternative mitigation prompt on Truthful / Persona / Alice-Bob / Snowballing settings. We can see that HK^+ mitigation is significantly higher than HK^-.

specific settings used (e.g., Alice-Bob, Truthful, etc.).

Figure 9 presents additional results for configurations beyond those discussed in the main paper. Specifically, Figures 9a and 9b illustrate the similarity of settings under the Gemma and Metta models, while Figures 9c and 9d highlight the hallucination similarity across models using Alice-Bob and Truthful settings. These findings mirror the patterns observed in the main paper and further underscore the differences between models and settings. This emphasizes the necessity of developing model- and setting-specific datasets.

G HK^+ Detection Additional Results

To complete the results from Section 6.1 on TriviaQA, we show in Figure 10 similar detection results that compare generic and model's specific dataset on Natural Questions.

H Preemptive Hallucination Detection Additional Results

In Section 6.2, we demonstrate that hallucinations can be detected before they are generated, whereas post-hallucination settings are less effective for this purpose. While the main paper presents results using the Mistral model, we extend this evaluation to Llama (Figure 11) and Gemma (Figure 12). The results for these models exhibit similar patterns to those observed with Mistral, reinforcing the generality of this phenomenon.

I Detection results on the MLP and Attention Components

The results in the main paper are only shown using hidden states from the residual component of the LLMs, that is, the representations after each transformer block. To complete the picture, we provide detection results also for the MLP and attention components using the representations that are output by the component.

The results are shown in Figures 13 and 14 for the classification into the two hallucination types and factually correct examples, for all the combinations of datasets and models.

Next in Figures 15 and 16 we see the results of the generalization of the Snowballing setting to the Alice-Bob setting using the MLP and Attention components.

Lastly, Figures 17 and 18 show detection of HK^+ results using classifiers trained on specific and generic datasets.

In all these figures, the results with the MLP and attention components yield similar trends to the ones in the main paper using the residual component, albeit with moderately lower accuracy. This implies that the detection results are not limited to a specific component and are a broader phenomenon across components.

J Generalization between Settings

In Section 5.2, we demonstrated that the Snowballing setting is generalizable to the Alice-Bob setting. To further validate this generalization, we present Figures 19 and 20, which show that Snowballing is also generalizable to the Persona and Truthful settings.

K Different snowballing Configuration have Similar Results

In the main paper, we used a single configuration with 3 bad shots, randomly selected. To demonstrate that our results are not dependent on this specific configuration, we conducted additional experiments with a different Snowballing setting. Specifically, we increased the number of shots to five and used a different random seed in the dataset creation process to sample differ-